

Second Mid Semester Examination
Cloud Computing-II
Branch: CSE-Semester: VI

Time 1:30 Hrs

Max. Marks 20

Q.1

- a) List any four key tools used in CI/CD pipelines in DevOps.
 - b) Explain how Jenkins integrates with Git and GitHub for Continuous Integration.
- (CO3 – 7 marks)**

OR

- a) A developer pushes faulty code that breaks the production environment. Based on DevOps practices, how should CI/CD tools handle such a situation automatically?
- b) Analyze the role of **repository management** and **code review tools like Gerrit** in DevOps pipelines. What challenges arise without them?

Q.2

- a) Name the components of **Chef Automate workflow** used for Continuous Delivery.
 - b) Explain the probative value of configuration management tools like **Ansible** in improving deployment consistency.
- (CO4 – 7 marks)**

OR

- a) How would **Ansible Vault** and **Ansible Tower** help in managing sensitive deployment credentials and large-scale automation?
- b) Compare the usage of **Splunk** and **Nagios** for monitoring DevOps environments. Which scenarios suit each best?

Q.3

- a) Define **Docker containerization** and name two advantages of using Docker.
 - b) Describe how **Kubernetes** differs from Docker in orchestrating containers.
- (CO5 – 6 marks)**

OR

- a) You are deploying a multi-container application. Explain two tools or practices to monitor and scale this system using Kubernetes.
- b) Analyze the role of **Helm** in managing Kubernetes applications. How does it improve productivity in large-scale deployments?